Andhra University College of Engineering

Civil Engineering | Course Code: CE301

Structural Analysis - Indeterminate Structural Analysis & Energy Methods Ma

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Overview: Deflection of determinate and indeterminate frames, Moment distribution iteration, Ka

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Deflection & Strain Energy Methods
Castigliano first and second theorems, Unit load method, Virtual work method for trusses and beams

Unit 2: Slope Deflection Method
Derivation of slope deflection equations, Analysis of continuous beams with sinking of supports, Rig

Unit 3: Moment Distribution Method
Stiffness factors, Carry-over factors, Distribution factors, Analysis of continuous beams and frames

Unit 4: Moving Loads & Influence Line Diagrams
Influence lines for bending moments and shear forces in simple beams, Muller-Breslau principle

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Analyze a continuous two-span beam with fixed ends subjected to uniformly distributed loads using Mome
2. Apply Castigliano second theorem to calculate vertical and horizontal deflections at the free end of a
3. Draw the influence line diagram for shear force and bending moment at a quarter-span section of a simp

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* C.S. Reddy - Basic Structural Analysis Tata McGraw-Hill 3rd Ed
* R.C. Hibbeler - Structural Analysis Pearson 10th Ed

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